

Elimination with matrices

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Any system of linear equations can be solved using elimination with matrices, given the system is consistent.

Suppose we have the system:

$$\begin{aligned} X + 2Y + Z &= 2 \\ 3X + 8Y + Z &= 12 \end{aligned}$$

$$4Y + Z = 2$$

We call this system $A\vec{X} = \vec{b}$, such that A is the coefficient matrix of the system, $\vec{X}$ is a vector of unknown scalars, and $\vec{b}$ is the right hand side of the equations, the constants if you like.

Think of $A\vec{X} = \vec{b}$ like:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$A_{n \times m}$ $X_{m \times 1}$ $b_{n \times 1}$

This means that if the system is solvable, i.e. the system is consistent, there exists some scalars, $x_1 \cdots x_n$, which when the matrix is multiplied by, will give $\vec{b}$.

Given the example at hand, we can write:

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}$$

Through matrix operations we will determine if there exist some x, y and z that will give $\vec{b}$.

Elimination with matrices (cont'd)

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Let's put the system into an augmented matrix and start reduction

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}$$

$x \quad y \quad z$ $x \quad y \quad z$

Augmented matrix

$$[A | \vec{b}] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array} \right]$$

$$\begin{aligned} R_2 &\leftarrow R_2 - 3R_1 \\ R_3 &\leftarrow R_3 - 2R_2 \end{aligned}$$

$$[U | \vec{c}] = \left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{2} & -2 & 6 \\ 0 & 0 & \boxed{5} & -10 \end{array} \right]$$

We now have the matrix in the form of U , and it's called Upper Triangular form.

We have the three pivots, all zeros underneath the pivots, and the pivot on the lower row lies to the right of the pivot in the row above.

We did no scaling or interchange of rows, meaning we can find the determinant of A as the product of the pivots

$$\det(A) = 1 \cdot 2 \cdot 5 = 10 \Rightarrow \neq 0, A \text{ is invertible.}$$

This matrix solved itself quite nicely, we have three pivots and can on basis of that tell that the system has a solution.

But, how could it have failed? How could we fail?

To come up with three pivots?

What if the last row of A was $[0 \ y \ -y]$? Then

we would end up with the last row of

$$[U | \vec{c}] = [0 \ 0 \ 0 | -10], \text{ suggesting } 0 = -10 \text{ which is impossible.}$$

Elimination with matrices (cont'd)

Having found $\vec{c} = \begin{bmatrix} 2 \\ 6 \\ -10 \end{bmatrix}$, we can back substitute into U to solve for $\vec{x}$.

We get:

$$\begin{aligned} 5z &= -10 \Rightarrow \boxed{z = -2} \\ 2y - 2z &= 6 \Rightarrow \boxed{y = 1} \\ x + 2y + z &= 2 \Rightarrow \boxed{x = 2} \end{aligned}$$

$\underbrace{\begin{bmatrix} z = -2 \\ y = 1 \\ x = 2 \end{bmatrix}}_{\vec{x}}$

Having found $\vec{x} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$, we can multiply

the matrix A by $\vec{x}$ to get $\vec{b}$.

$$\begin{aligned} A\vec{x} &= \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 + 1 \cdot (-2) \\ 3 \cdot 2 + 8 \cdot 1 + 1 \cdot (-2) \\ 0 \cdot 2 + 4 \cdot 1 + 1 \cdot (-2) \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix} = \vec{b}$$

Having done this exercise, we see that we can look at the columns in A as vectors which we can scale using the entries in $\vec{x}$, and add to get to $\vec{b}$. We can write:

$$2 \cdot \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 2 \\ 8 \\ 4 \end{bmatrix} - 2 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix} = \vec{b}$$